



BITS Pilani
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Natural Language Processing Lab 1

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In this Lab Demo we will cover:

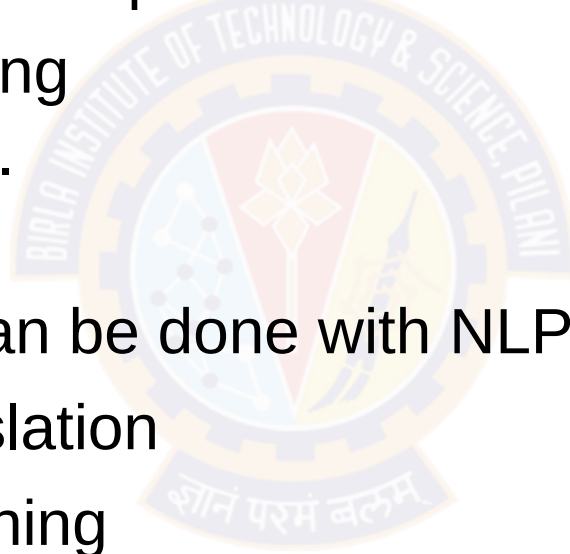
- What is Natural Language Processing (NLP)?
- NLP Applications
- Natural language Processing Tools
- Natural Language Toolkit (NLTK)
 - Installation
 - Tokenization word tokenizer and sentence tokenizer
 - Stemming -Porter Stemmer
 - Lemmatization

What is Natural Language Processing?

- Analyze, understand and generate human languages just like humans do.
- Applying computational techniques to language domain.
- To explain linguistic theories, to use the theories to build systems that can be of social use.
- Started off as a branch of Artificial Intelligence.
- Borrows from Linguistics, Psycholinguistics, Cognitive Science & Statistics.
- Make computers learn our language rather than we learn theirs.

NLP Applications

- Question answering
 - Who is the first Taiwanese president?
- Text Categorization/Routing
 - e.g., customer e-mails.
- Text Mining
 - Find everything that can be done with NLP
- Machine (Assisted) Translation
- Language Teaching/Learning
 - Usage checking
- Spelling correction
 - Is that just dictionary lookup?



NLP Tools

Some commercial tools

- [IBM Watson](#) | A pioneer AI platform for businesses
- [Google Cloud NLP API](#) | Google technology applied to NLP
- [Amazon Comprehend](#) | An AWS service to get insights from text

Open Source Tools

- [Stanford Core NLP](#) is a popular Java library built and maintained by Stanford University.
- **SpaCy** - One of the newest open-source Natural Language Processing with Python libraries
- [Gensim](#) is a highly specialized Python library that largely deals with topic modeling tasks using algorithms like Latent Dirichlet Allocation (LDA)
- [Natural Language Toolkit \(NLTK\)](#) is the most popular Python library

Python, PIP and Anaconda Installation links

Python installation

<https://www.python.org/downloads/>

Link for PIP installation on windows/LINUS/MAC

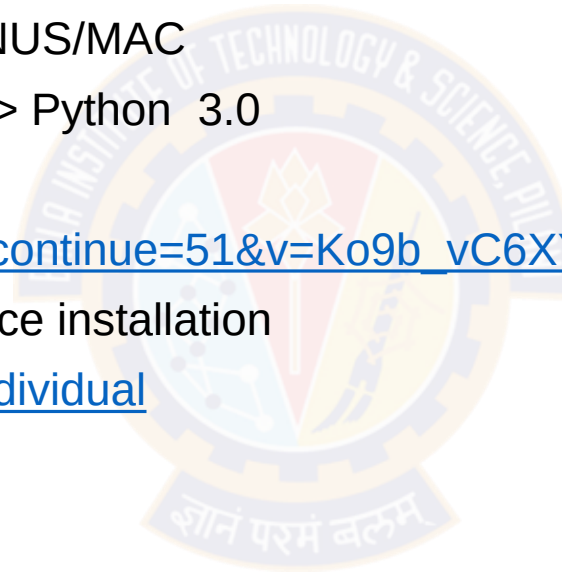
PIP is already installed for all versions > Python 3.0

<https://pip.pypa.io/en/stable/installing/>

https://www.youtube.com/watch?time_continue=51&v=Ko9b_vC6XY0&feature=emb_logo

Link for Anaconda individual open source installation

<https://www.anaconda.com/products/individual>



NLTK installation

- The Natural Language Toolkit (NLTK) is a platform used for building programs for text analysis.

Open Anaconda terminal, run

pip install nltk.

Anaconda and Jupiter are best and popular data science tools

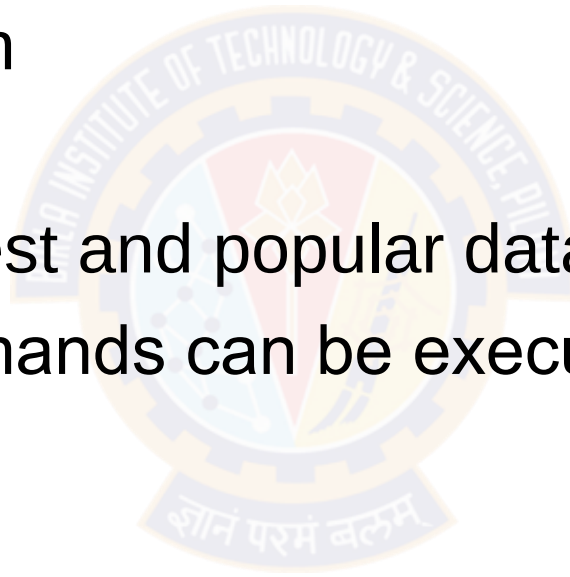
In Jupyter, the console commands can be executed by the '!' sign before the command within the cell.

! pip install nltk

[NLTK book](#)

[NLTK discussion forum](#)

<https://www.nltk.org/install.html>



NLP Lab Demo

- Installation
- Loading the data
- Understand the data
- Data Pre-processing
 - ☐ Tokenization
 - ☐ Remove Punctuations
 - ☐ Remove Stopwords
 - ☐ Stemming/Lemmatization



Summary

- We've seen an important text preprocessing techniques like tokenization, stop word removal, stemming and lemmatization in NLTK





Thank You!